

Arithmetic Spectra of the Rvachev Sinc Product

Thue–Morse lobe signs, exact lattice reproduction, complex dimensions, and finite q -binomial spectral layering

Frontier report prepared for Vladimir Reshetnikov
after a recursive audit of the complete ProveIt Fabius-function TeX corpus

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Abstract

This report starts from every `*.tex` document under `Analysis/FabiusFunction/docs` at the frozen commit named above. The corpus contains the principal Fabius–Rvachev exposition, its historical revisions, a Lean walkthrough and glossary, exact dyadic and q -calculus studies, repeated-integration and spline reports, endpoint asymptotics, a large Thue–Morse formula atlas, a sequence of Fourier-decay audits, and primary or transcribed papers. Repeated revisions were read as one mathematical lineage; the latest corrected statement was used whenever sibling drafts disagreed.

The principal synthesis is that the canonical-product zero multiplicities of the Fourier transform

$$\Phi(z) = \widehat{\text{up}}(z) = \prod_{h \geq 0} \text{sinc}\left(\frac{\pi z}{2^h}\right) = \prod_{n \geq 1} \left(1 - \frac{z^2}{n^2}\right)^{1+\nu_2(n)}$$

form an arithmetic spectrum. Its multiplicity function $a(n) = 1 + \nu_2(n)$ has Dirichlet series $\zeta(s)/(1 - 2^{-s})$, exact counting law $\sum_{n \leq N} a(n) = 2N - s_2(N)$, and a dense dyadic natural boundary in its ordinary generating function. These elementary facts organize several results which do not appear as a theorem package in the pinned corpus.

First, the sign on the N th Fourier lobe is exactly the signed Thue–Morse bit $(-1)^{s_2(N)}$. This pattern had been observed before; here it is made a direct corollary of the exact zero count and is converted into an absolutely convergent Thue–Morse lobe decomposition of up and all its derivatives. Second, the zero multiplicity gives an exact Strang–Fix hierarchy: on the lattice $N^{-1}\mathbb{Z}$, polynomially weighted periodizations are constant through degree $\nu_2(N)$, and the next degree fails. A reciprocal-moment Appell correction then reproduces every polynomial of that degree exactly. Thus the approximation order of the translate system is controlled not by the size of N , but by its 2-adic valuation.

Third, the zero spectrum has a spectral zeta function with “complex dimensions” $2\pi i k / \log 2$. Mellin inversion gives a superalgebraically accurate expansion of $\log \Phi(iy)$ with the same Gamma–zeta Fourier coefficients that occur in the repository’s corrected endpoint asymptotics. This supplies a new structural explanation of that periodic fluctuation: it is the residue trace of the dyadic zero spectrum. Finite sinc products replace the geometric denominator by the q -integer $[J]_{2^{-s}}$; the imaginary-axis poles are absent at every finite stage and emerge through an explicit pole-condensation limit. Gaussian q -binomial coefficients appear as the algebra of independently selecting spectral layers, while the common-zero statistics form a hierarchy

$$\sum_{n \geq 1} \binom{1 + \nu_2(n)}{r} n^{-s} = \zeta(s) \frac{2^{-(r-1)s}}{(1 - 2^{-s})^r}.$$

A heat-trace expansion and a base- b extension complete the report.

Claims marked NEW-TO-APSHOT mean only that no counterpart was found in the frozen TeX corpus after targeted collision searches. They are proved here, but no claim of worldwide historical priority is made. In particular, Rvachev’s 1977 paper on approximation by up and the broader Russian-language atomic-function literature require a separate priority audit.

Status convention. CORPUS denotes an input already developed in the repository or cited primary literature. DERIVED denotes a short consequence included to make the argument self-contained. NEW-TO-APSHOT denotes a theorem not located in the pinned TeX snapshot. PRIOR OBSERVATION denotes a phenomenon recorded before but sharpened or reorganized here. None of the new-to-snapshot results is represented as already formalized in Lean.

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1 Audit boundary and research method

1.1 Why the snapshot is pinned

The default branch changed while this audit was in progress. To prevent a moving target, all corpus statements in this report refer to commit `0b1db2f91321b6575be9a712dc1025e88af318db`. Its commit time is 27 August 2026, 18:19:55 UTC. GitHub’s code index reported 54 TeX files under `Analysis/FabiusFunction/docs` at audit time; because large generated and consolidated sources are not always returned reliably by text search, the recursive Git tree, rather than the index, was used as the authoritative inventory.

The documents fall into the following overlapping families.

Family	Mathematical role
Primary exposition	Definitions, probability model, functional equations, canonical sinc product, exact dyadic values, derivative identities, partition of unity, and the corrected global asymptotic framework.
Revision archive	Earlier expositions, supplements, glossaries, and walkthroughs; useful for detecting abandoned normalizations and claims corrected in later waves.
Dyadic and q -calculus	Moment recurrences, Appell blocks, Gaussian-binomial stencils, q -Pochhammer normalizations, exact dyadic arithmetic, Toeplitz limits, and accelerated spline approximants.
Repeated integration	Finite nonuniform box splines, exact Thue–Morse truncated powers, Peano kernels, and sharp 4^{-n} convergence laws.
Endpoint asymptotics	Correct Lambert phase, all-order saddle expansion, explicit low-order coefficients, and the Gamma–zeta Fourier law for the log-periodic term.
Thue–Morse atlases	Digital, valuative, Pascal, automata, Mahler, finite-difference, rarefaction, Mellin–Dirichlet, word-combinatorial, and spectral representations.
Fourier-decay audits	Successive corrections separating geometric, arithmetic, RMS, and envelope exponents for the real-axis infinite sinc product.
Primary papers	Arias de Reyna’s analytic and arithmetic papers, lacunary-product work, Kronecker-sequence work, and OCR-corrected or transcribed source material.

1.2 Collision policy

A formula is not called new merely because it admits a different notation. The following attractive directions were deliberately excluded from the novelty claims.

- The all-order endpoint coefficients, including explicit A_2 and A_3 , are already in the repository’s small-argument asymptotics study.
- Moment deconvolution and Richardson acceleration already occur in archived dyadic and convergence reports, including a half-base Gaussian-binomial filter across spline degrees.
- Exact zero multiplicity $1 + \nu_2(n)$ is already formalized in the Lean canonical-product module.
- The sign pattern of the Fourier lobes was stated in Vladimir Reshetnikov’s 2018 Math Stack-Exchange question, and the answer there essentially proves uniqueness of the critical point in every lobe from the logarithmic derivative.

- Ordinary partition of unity is classical for up and appears in the primary literature. The new-to-snapshot statement below is the sharp higher weighted hierarchy and its exact 2-adic stopping order.

The strategy was therefore to promote the zero multiplicity from an isolated canonical-product fact to a spectrum and ask what its standard transforms know. The answer simultaneously touches Dirichlet series, Mellin residues, automatic sequences, shift-invariant approximation, and finite q -calculus.

2 Normalization and established inputs

2.1 The centered random series

Let $U_1, U_2, \dots$ be independent and uniform on $[-1, 1]$, and set

$$Y = \sum_{k=1}^{\infty} 2^{-k} U_k. \quad (2.1)$$

Then Y is supported on $[-1, 1]$ and has the even density up. Under the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx, \quad (2.2)$$

independence gives

$$\Phi(\xi) := \widehat{\text{up}}(\xi) = \prod_{h=0}^{\infty} \text{sinc}\left(\frac{\pi \xi}{2^h}\right), \quad \text{sinc } z = \frac{\sin z}{z}. \quad (2.3)$$

The convergence is locally uniform on $\mathbb{C}$, so Φ is an even entire function. The density is C^∞ and compactly supported; consequently Φ is rapidly decreasing on the real axis, although its sharp pointwise behavior is highly nonuniform.

The bounded Fabius variable is

$$X = \frac{Y+1}{2} = \sum_{k=1}^{\infty} 2^{-k} V_k, \quad V_k \sim \text{Unif}[0, 1], \quad (2.4)$$

with distribution function F . This affine relation will be the bridge between the imaginary-axis growth of Φ and the small-argument asymptotics of F .

2.2 Canonical product and exact orders

Euler's product for sine and an absolutely convergent rearrangement give

$$\Phi(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)^{a(n)}, \quad a(n) = 1 + \nu_2(n). \quad (2.5)$$

Thus

$$\text{ord}_{z=n} \Phi = a(n), \quad n \in \mathbb{Z} \setminus \{0\}. \quad (2.6)$$

This identity is an established corpus input. Everything in [sections 3, 6, 8 and 9](#) is driven by taking it seriously as spectral data rather than merely as a description of zeros.

The first centered moments needed later are

$$\mu_0 = 1, \quad \mu_1 = 0, \quad \mu_2 = \frac{1}{9}, \quad \mu_3 = 0, \quad \mu_4 = \frac{19}{675}, \quad (2.7)$$

where $\mu_r = \int x^r \text{up}(x) dx$. They follow either from the random series or from the repository's moment recurrence.

3 The arithmetic zero spectrum

3.1 Equivalent descriptions

Theorem 3.1 (Arithmetic spectrum package NEW-TO-SNAPSHOT). *For $n \geq 1$, let $a(n) = 1 + \nu_2(n)$. Then:*

$$a(n) = \sum_{h \geq 0} \mathbf{1}_{2^h | n}, \quad (3.1)$$

$$\sum_{n \geq 1} \frac{a(n)}{n^s} = \frac{\zeta(s)}{1 - 2^{-s}}, \quad \Re s > 1, \quad (3.2)$$

$$A(N) := \sum_{n \leq N} a(n) = \sum_{h \geq 0} \left\lfloor \frac{N}{2^h} \right\rfloor = 2N - s_2(N), \quad (3.3)$$

$$\#\{n \leq N : a(n) \geq r\} = \left\lfloor \frac{N}{2^{r-1}} \right\rfloor, \quad (3.4)$$

$$\#\{n \leq N : a(n) = r\} = \left\lfloor \frac{N}{2^{r-1}} \right\rfloor - \left\lfloor \frac{N}{2^r} \right\rfloor. \quad (3.5)$$

Moreover a is multiplicative and satisfies

$$a(2n) = a(n) + 1, \quad a(2n + 1) = 1. \quad (3.6)$$

Its limiting value distribution is geometric:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \#\{n \leq N : a(n) = r\} = 2^{-r}, \quad r \geq 1, \quad (3.7)$$

so the limiting mean and variance are both 2.

Proof. The number of nonnegative h for which $2^h \mid n$ is exactly $1 + \nu_2(n)$, proving (3.1). Absolute convergence permits summation by layers:

$$\sum_{n \geq 1} \frac{a(n)}{n^s} = \sum_{h \geq 0} \sum_{2^h | n} n^{-s} = \zeta(s) \sum_{h \geq 0} 2^{-hs}.$$

The same layer count truncated at N gives the first expression for $A(N)$. Legendre’s digit-sum identity $\sum_{h \geq 1} \lfloor N/2^h \rfloor = N - s_2(N)$ gives the last one. The inequalities $a(n) \geq r$ and $2^{r-1} \mid n$ are equivalent, giving (3.4); subtraction gives (3.5). Multiplicativity follows because only the 2-part contributes: if $(m, n) = 1$, at most one is even, and $a(mn) = a(m)a(n)$. The recursion and limiting law are immediate. The geometric mean is $\sum r 2^{-r} = 2$ and its variance is 2. $\square$

Remark 3.2. The average zero multiplicity is therefore 2, while exceptional powers of two have unbounded multiplicity. This is the first indication that average real-axis behavior and extremal dyadic behavior should not be conflated—precisely the lesson reached independently by the repository’s Fourier-decay audit sequence.

3.2 Ordinary generating function and natural boundary

Proposition 3.3 (Lambert series and a dyadic natural boundary NEW-TO-SNAPSHOT). *For $|z| < 1$,*

$$\mathcal{A}(z) := \sum_{n \geq 1} a(n) z^n = \sum_{h \geq 0} \frac{z^{2^h}}{1 - z^{2^h}}, \quad (3.8)$$

with Mahler recursion

$$\mathcal{A}(z) = \frac{z}{1-z} + \mathcal{A}(z^2). \quad (3.9)$$

Every root of unity of order a power of two is a radial singularity. Since these roots are dense on the unit circle, $|z| = 1$ is a natural boundary.

Proof. Expand (3.1) and sum geometric series. Splitting off $h = 0$ and shifting the remaining index gives (3.9). If $\omega^{2^m} = 1$ and $z = r\omega$ with $r \uparrow 1$, the $h = m$ summand is $r^{2^m}/(1 - r^{2^m}) \rightarrow +\infty$. Thus analytic continuation across any arc is impossible. $\square$

3.3 Spectral traces and the Taylor law

Corollary 3.4 (Even spectral traces NEW-TO-SNAPSHOT). *For every integer $m \geq 1$,*

$$\sum_{n \geq 1} \frac{a(n)}{n^{2m}} = \frac{\zeta(2m)}{1 - 4^{-m}}, \quad (3.10)$$

and, for $|z| < 1$,

$$\log \Phi(z) = - \sum_{m=1}^{\infty} \frac{\zeta(2m)}{m(1 - 4^{-m})} z^{2m}. \quad (3.11)$$

Proof. The first formula is (3.2) at $s = 2m$. Expanding $\log(1 - z^2/n^2)$ in the canonical product and interchanging the absolutely convergent sums gives the second. $\square$

The logarithmic derivative is the corresponding resolvent trace:

$$\frac{\Phi'(z)}{\Phi(z)} = \sum_{n \geq 1} a(n) \left(\frac{1}{z-n} + \frac{1}{z+n} \right) = -2z \sum_{n \geq 1} \frac{a(n)}{n^2 - z^2}. \quad (3.12)$$

This identity will control the geometry of every real lobe.

4 Thue–Morse signs from zero counting

4.1 The neighboring-bit identity

Let

$$\varepsilon_n = (-1)^{s_2(n)}, \quad n \geq 0, \quad (4.1)$$

be the signed Thue–Morse sequence.

Lemma 4.1 (Multiplicity parity is a Thue–Morse discrete derivative). *For every $n \geq 1$,*

$$(-1)^{a(n)} = \varepsilon_{n-1} \varepsilon_n. \quad (4.2)$$

Proof. Adding one to $n - 1$ erases $\nu_2(n)$ trailing ones and creates one new one, so $s_2(n) = s_2(n - 1) + 1 - \nu_2(n)$. Hence $s_2(n - 1) + s_2(n) \equiv 1 + \nu_2(n) = a(n) \pmod{2}$. $\square$

4.2 Exact lobe signs

Theorem 4.2 (Thue–Morse lobe-sign theorem PRIOR OBSERVATION). *For real $x \notin \mathbb{Z}$,*

$$\operatorname{sgn} \Phi(x) = (-1)^{s_2(\lfloor |x| \rfloor)} = \varepsilon_{\lfloor |x| \rfloor}. \quad (4.3)$$

Equivalently, the signs of the consecutive positive-axis lobes of the infinite sinc product are exactly the signed Thue–Morse word.

Proof. The product is positive at the origin. Crossing the zero at n reverses sign exactly when $a(n)$ is odd. Therefore, for $N < x < N + 1$,

$$\operatorname{sgn} \Phi(x) = (-1)^{A(N)}.$$

By (3.3), $A(N) = 2N - s_2(N)$, whose parity is $s_2(N)$. Evenness handles negative x . $\square$

Remark 4.3. The pattern itself predates this report. The point of [theorem 4.2](#) is the exact spectral explanation: Thue–Morse is not an external decoration placed on the sinc product. It is the parity integral of its 2-adic zero multiplicities, as (4.2) makes literal.

For the J -factor truncation

$$\Phi_J(x) = \prod_{h=0}^{J-1} \operatorname{sinc}\left(\frac{\pi x}{2^h}\right), \quad (4.4)$$

we likewise have, away from its zeros,

$$\operatorname{sgn} \Phi_J(x) = (-1)^{\sum_{h=0}^{J-1} \lfloor |x|/2^h \rfloor}. \quad (4.5)$$

Thus the infinite Thue–Morse sign law is already visible through finite digital floor sums.

4.3 A Thue–Morse lobe decomposition of the bump

Theorem 4.4 (Signed lobe inversion NEW-TO-SPASHOT). *For every real x ,*

$$\operatorname{up}(x) = 2 \sum_{N=0}^{\infty} \varepsilon_N \int_N^{N+1} |\Phi(\xi)| \cos(2\pi x \xi) \, d\xi. \quad (4.6)$$

The series and every termwise differentiated series converge absolutely. In particular, for $m \geq 0$,

$$\operatorname{up}^{(2m)}(0) = 2(-1)^m (2\pi)^{2m} \sum_{N=0}^{\infty} \varepsilon_N \int_N^{N+1} \xi^{2m} |\Phi(\xi)| \, d\xi. \quad (4.7)$$

For $m = 0$ this becomes the exact alternating-area identity

$$1 = 2 \sum_{N \geq 0} \varepsilon_N \int_N^{N+1} |\Phi(\xi)| \, d\xi. \quad (4.8)$$

Proof. Fourier inversion and evenness give $\operatorname{up}(x) = 2 \int_0^{\infty} \Phi(\xi) \cos(2\pi x \xi) \, d\xi$. Partition the half-line into unit lobes and apply [theorem 4.2](#). Rapid decay of Φ permits arbitrary polynomial weights, proving absolute convergence and termwise differentiation. $\square$

This identity is a genuinely analytic bridge back from the sinc product to the Thue–Morse sequence. It differs from the better-known appearance of Thue–Morse in truncated-power formulas for finite box splines: here the sequence weights frequency lobes of the limiting C^∞ bump itself.

5 Laguerre–Polya geometry and real critical points

Theorem 5.1 (Real-rooted derivative hierarchy DERIVED). *The entire function Φ belongs to the Laguerre–Polya class. Consequently, every derivative $\Phi^{(r)}$ has only real zeros. More precisely:*

1. *the zero $n \in \mathbb{Z} \setminus \{0\}$ is inherited by Φ' with multiplicity $a(n) - 1$;*
2. *in every interval $(n, n + 1)$, $n \geq 0$, Φ' has exactly one simple zero c_n ;*

3. for every $r \geq 0$ and nonzero integer m ,

$$\Phi^{(r)}(2^r m) = 0. \quad (5.1)$$

Proof. The canonical product is the locally uniform limit of real-rooted even polynomials, because $\sum_n a(n)n^{-2} < \infty$. Hence it is in the Laguerre–Polya class, which is closed under differentiation.

An order- $a(n)$ zero loses one order under differentiation. On $(n, n+1)$, (3.12) is strictly decreasing, since

$$\left(\frac{\Phi'}{\Phi}\right)'(x) = -\sum_{m \geq 1} a(m) \left(\frac{1}{(x-m)^2} + \frac{1}{(x+m)^2} \right) < 0.$$

It tends to $+\infty$ at $n+$ and to $-\infty$ at $(n+1)-$, so it has one simple zero. Finally, the zero at $2^r m$ has multiplicity at least $r+1$. $\square$

Corollary 5.2 (Signed unique extrema). *The unique critical value in the n th lobe has sign ε_n . Thus the critical-value sequence may be written*

$$\Phi(c_n) = \varepsilon_n M_n, \quad M_n > 0. \quad (5.2)$$

The repository’s Fourier-decay program studies the highly irregular envelope M_n . The arithmetic spectrum explains why a single “typical” exponent cannot capture all subsequences: the inherited multiplicity at each endpoint varies with $\nu_2(n)$, while the lobe sign varies with the integrated parity $s_2(n) \bmod 2$.

6 Exact 2-adic Strang–Fix hierarchy

This section contains the strongest new-to-snapshot theorem. The ordinary partition of unity is the degree-zero case. The full hierarchy reveals an exact and sharp dependence on the 2-adic valuation of the lattice denominator.

6.1 Weighted periodizations

For $N \geq 1$ and $r \geq 0$, define

$$\mathcal{P}_{N,r}(x) = \sum_{k \in \mathbb{Z}} \left(x - \frac{k}{N}\right)^r \text{up}\left(x - \frac{k}{N}\right). \quad (6.1)$$

The sum is locally finite because up has compact support.

Theorem 6.1 (Sharp weighted partition hierarchy NEW-TO-SNAPSHOT). *Let $q = \nu_2(N)$. For every $0 \leq r \leq q$ and every real x ,*

$$\boxed{\mathcal{P}_{N,r}(x) = N\mu_r}. \quad (6.2)$$

The statement is sharp: $\mathcal{P}_{N,q+1}$ is not constant. Equivalently, the translate system generated by up on the lattice $N^{-1}\mathbb{Z}$ has exact Strang–Fix order

$$\text{SF}(N) = 1 + \nu_2(N). \quad (6.3)$$

Proof. Put $g_r(t) = t^r \text{up}(t)$. Poisson summation for the N^{-1} -periodization gives

$$\mathcal{P}_{N,r}(x) = N \sum_{m \in \mathbb{Z}} \widehat{g}_r(mN) e^{2\pi i m N x}. \quad (6.4)$$

Differentiating the Fourier transform,

$$\widehat{g}_r(\xi) = \frac{\Phi^{(r)}(\xi)}{(-2\pi i)^r}. \quad (6.5)$$

For $m \neq 0$, the zero at mN has order

$$a(mN) = 1 + \nu_2(mN) = 1 + q + \nu_2(m) \geq q + 1.$$

Hence $\Phi^{(r)}(mN) = 0$ for $r \leq q$. Only $m = 0$ remains in (6.4), giving $N\widehat{g}_r(0) = N\mu_r$. At $r = q + 1$, the Fourier coefficient with $m = 1$ is nonzero because the zero at N has exact order $q + 1$. Therefore the periodization cannot be constant. $\square$

Example 6.2 (The first three layers). For every N ,

$$\sum_{k \in \mathbb{Z}} \text{up}(x - k/N) = N. \quad (6.6)$$

If $2 \mid N$, then

$$\sum_{k \in \mathbb{Z}} (x - k/N) \text{up}(x - k/N) = 0. \quad (6.7)$$

If $4 \mid N$, then

$$\sum_{k \in \mathbb{Z}} (x - k/N)^2 \text{up}(x - k/N) = \frac{N}{9}. \quad (6.8)$$

The quadratic identity fails for $N \equiv 2 \pmod{4}$ and the linear identity fails for odd N .

Remark 6.3 (Why the result is arithmetically unusual). Refining the lattice from $1/N$ to $1/(N + 1)$ can reduce the exact reproduction order dramatically. Powers of two are optimal: the lattice $2^{-q}\mathbb{Z}$ has order $q + 1$, whereas every odd denominator has only the classical order one.

6.2 Exact polynomial reproduction

Let

$$M(z) = \int_{\mathbb{R}} e^{zt} \text{up}(t) dt = \sum_{r \geq 0} \mu_r \frac{z^r}{r!}. \quad (6.9)$$

Since up is even, $M(-z) = M(z)$. Define reciprocal-moment Appell polynomials P_r by

$$\frac{e^{xz}}{M(z)} = \sum_{r=0}^{\infty} P_r(x) \frac{z^r}{r!}. \quad (6.10)$$

Theorem 6.4 (Appell-corrected lattice reproduction NEW-TO-SNAPSHOT). *If $q = \nu_2(N)$, then for every polynomial degree $0 \leq r \leq q$,*

$$\boxed{\frac{1}{N} \sum_{k \in \mathbb{Z}} P_r(k/N) \text{up}(x - k/N) = x^r.} \quad (6.11)$$

Consequently every polynomial of degree at most q is reproduced exactly by a locally finite linear combination of N^{-1} -lattice translates of up .

Proof. Write $t_k = x - k/N$. Up to degree q , [theorem 6.1](#) implies the formal Taylor identity

$$\frac{1}{N} \sum_k e^{-zt_k} \text{up}(t_k) = M(-z) + \mathcal{O}(z^{q+1}) = M(z) + \mathcal{O}(z^{q+1}).$$

Multiplying by $e^{xz}/M(z)$ yields

$$\frac{1}{N} \sum_k \frac{e^{(x-t_k)z}}{M(z)} \text{up}(t_k) = e^{xz} + \mathcal{O}(z^{q+1}).$$

Comparing coefficients gives (6.11). $\square$

The first correction polynomials are

$$\begin{aligned} P_0(x) &= 1, \\ P_1(x) &= x, \\ P_2(x) &= x^2 - \frac{1}{9}, \\ P_3(x) &= x^3 - \frac{x}{3}, \\ P_4(x) &= x^4 - \frac{2}{3}x^2 + \frac{31}{675}. \end{aligned} \tag{6.12}$$

Thus, for example, the $1/16$ lattice reproduces quartics exactly.

6.3 A formalization-friendly version

The analytic proof above uses Poisson summation, but the theorem can be decomposed for Lean into three reusable statements:

1. Fourier transform of $x^r \text{up}(x)$ equals $(-2\pi i)^{-r} \Phi^{(r)}$;
2. periodization on $N^{-1}\mathbb{Z}$ has Fourier coefficients sampled at mN ;
3. exact zero order at mN is $1 + \nu_2(mN)$.

No asymptotic estimate is needed. Once a suitable Poisson theorem for compactly supported smooth functions is available, the sharpness proof is a one-coefficient argument.

7 Spectral zeta function and complex dimensions

Definition 7.1 (Zero spectral zeta function). For $\Re s > 1$, set

$$Z_\Phi(s) = \sum_{n \geq 1} \frac{\text{ord}_n \Phi}{n^s} = \sum_{n \geq 1} \frac{a(n)}{n^s}. \tag{7.1}$$

By (3.2),

$$Z_\Phi(s) = \frac{\zeta(s)}{1 - 2^{-s}}. \tag{7.2}$$

It extends meromorphically to $\mathbb{C}$. Besides the ordinary density pole at $s = 1$, it has the dyadic lattice of poles

$$\chi_k = \frac{2\pi i k}{\log 2}, \quad k \in \mathbb{Z}, \tag{7.3}$$

except where an accidental zeta zero would cancel one. At $k \neq 0$,

$$\text{Res}_{s=\chi_k} Z_\Phi(s) = \frac{\zeta(\chi_k)}{\log 2}. \tag{7.4}$$

We call the χ_k the complex dimensions of the dyadic zero spectrum. This terminology is descriptive: the underlying object is simply the meromorphic Dirichlet series (7.2).

Remark 7.2. The pole at $s = 1$ has residue 2, agreeing with the exact zero count $A(N) = 2N - s_2(N)$. The imaginary lattice records the discrete scale invariance which is invisible in the leading density but reappears as a periodic function of $\log_2$ scale.

8 Mellin transform on the imaginary axis

8.1 Exact transform

Set

$$H(y) = \log \Phi(iy) = \sum_{n \geq 1} a(n) \log \left(1 + \frac{y^2}{n^2} \right), \quad y > 0. \quad (8.1)$$

All terms are positive, so H has none of the real-axis cancellation difficulties.

Theorem 8.1 (Mellin factorization NEW-TO-SPASHOT). *For $1 < \Re s < 2$,*

$$\int_0^\infty H(y) y^{-s-1} dy = \frac{\pi \zeta(s)}{s \sin(\pi s/2) (1 - 2^{-s})}. \quad (8.2)$$

Equivalently, with $S = 2\pi y$,

$$H(y) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\Gamma(1-s) \zeta(1-s)}{s(1-2^{-s})} S^s ds, \quad 1 < c < 2. \quad (8.3)$$

Proof. For $0 < \Re s < 2$, integration by parts and the beta integral give

$$\int_0^\infty \log(1+t^2) t^{-s-1} dt = \frac{\pi}{s \sin(\pi s/2)}. \quad (8.4)$$

For $\Re s > 1$, Tonelli's theorem and $y = nt$ factor the sum into this kernel and $Z_\Phi(s)$. The second form follows from the zeta functional equation. $\square$

8.2 Complete large- y expansion

Let $L = \log 2$ and define

$$C_2 = \text{rac} \zeta''(0) + \frac{1}{2} \log^2(2\pi) - \frac{\pi^2}{24} L - \frac{L}{12} = -1.1090453654341866821907455547 \dots \quad (8.5)$$

For $k \neq 0$, let

$$c_k = -\frac{\Gamma(-\chi_k) \zeta(1-\chi_k)}{L}, \quad (8.6)$$

and define the real analytic, mean-zero, 1-periodic function

$$\Psi(u) = \sum_{k \neq 0} c_k e^{2\pi i k u}. \quad (8.7)$$

Theorem 8.2 (Complex-dimension expansion NEW-TO-SPASHOT). *For every $A > 0$, as $y \rightarrow \infty$,*

$$H(y) = S - \frac{\log^2 S}{2L} - \frac{1}{2} \log S + C_2 + \Psi(\log_2 S) + \mathcal{O}_A(S^{-A}), \quad S = 2\pi y. \quad (8.8)$$

The Fourier coefficients decay exponentially. Numerically,

$$\begin{aligned} c_1 &= (7.5586509397640965 - 7.4700909117916216i) 10^{-7}, \\ |c_1| &= 1.0627109779221651 \times 10^{-6}, \\ |c_2| &= 6.6927771176343188 \times 10^{-13}. \end{aligned} \quad (8.9)$$

The exact triangle estimate is

$$\|\Psi\|_\infty \leq 2 \sum_{k \geq 1} |c_k|, \quad 2 \sum_{k \geq 1} |c_k| \approx 2.1254232944005 \times 10^{-6}, \quad (8.10)$$

where the displayed decimal is an 80-digit-working-precision numerical sum.

Proof architecture. Shift the contour in (8.3) to the left. The pole at $s = 1$ gives S . The triple pole at $s = 0$ gives the quadratic logarithm, linear logarithm, and C_2 . The simple poles at $s = \chi_k$, $k \neq 0$, give $c_k S^{\chi_k} = c_k e^{2\pi i k \log_2 S}$. In the functional-equation form, there are no further poles to the left: the trivial zeta zeros cancel the sine poles. Stirling’s estimate for $\Gamma(1 - s)$ permits a shift to any fixed negative real part, giving the superalgebraic remainder. The same estimate proves exponential Fourier decay. $\square$

8.3 Endpoint duality

Define the negative Laplace transform of the bounded Fabius variable by

$$\Lambda(S) = \mathbb{E} e^{-2SX}. \quad (8.11)$$

From $X = (Y + 1)/2$ and symmetry of Y ,

$$\boxed{\Phi(iy) = e^S \Lambda(S), \quad S = 2\pi y.} \quad (8.12)$$

Thus (8.8) is exactly the logarithmic large-parameter expansion which feeds the repository’s small-endpoint saddle point. The Gamma–zeta coefficients c_k already occur there, but [theorem 8.2](#) supplies a new spectral origin:

$$\text{endpoint log-periodicity} \quad \longleftrightarrow \quad \text{residues of } Z_\Phi(s) \text{ at } \chi_k. \quad (8.13)$$

The oscillation is tiny not because the dyadic scaling is weak, but because $\Gamma(-\chi_k)$ is exponentially small at the first nonzero imaginary height.

9 Finite products, q -integers, and pole condensation

9.1 Truncated multiplicities

The J -factor product (4.4) has multiplicity

$$a_J(n) = \min\{J, 1 + \nu_2(n)\}. \quad (9.1)$$

Let $q = 2^{-s}$. Then

Theorem 9.1 (Finite spectral q -integer NEW-TO-SNAPSHOT). *For $\Re s > 1$,*

$$Z_{\Phi_J}(s) := \sum_{n \geq 1} \frac{a_J(n)}{n^s} = \zeta(s) \frac{1 - 2^{-Js}}{1 - 2^{-s}} = \zeta(s) [J]_q, \quad (9.2)$$

where $[J]_q = 1 + q + \cdots + q^{J-1}$. Also

$$\sum_{n \leq N} a_J(n) = \sum_{h=0}^{J-1} \left\lfloor N/2^h \right\rfloor, \quad (9.3)$$

while

$$\log \Phi_J(z) = - \sum_{m \geq 1} \frac{\zeta(2m)}{m} [J]_{4^{-m}} z^{2m} \quad (|z| < 1). \quad (9.4)$$

Every finite Z_{Φ_J} is regular at the nonzero χ_k : the apparent geometric denominator is canceled by the numerator. The complex dimensions are therefore genuinely infinite-depth phenomena.

Theorem 9.2 (Local pole-condensation profile NEW-TO-SNAPSHOT). *Fix $k \in \mathbb{Z}$ and put $\chi_k = 2\pi i k / L$. Uniformly for w in compact subsets of $\mathbb{C}$,*

$$\lim_{J \rightarrow \infty} \frac{1}{J} Z_{\Phi_J} \left(\chi_k + \frac{w}{J} \right) = \zeta(\chi_k) \frac{1 - e^{-Lw}}{Lw}, \quad (9.5)$$

with the removable value 1 for the last factor at $w = 0$.

Proof. At $s = \chi_k + w/J$, one has $q^J = 2^{-Js} = e^{-Lw}$ and $J(1 - q) \rightarrow Lw$. Substitute in (9.2). $\square$

The profile $(1 - e^{-Lw})/(Lw)$ is the finite-depth regularization of the limiting pole. It gives a precise analytic meaning to the statement that a q -integer turns into a geometric resolvent.

9.2 Gaussian q -binomial layer algebra

Define the layer zeta functions

$$\mathcal{Z}_h(s) = \sum_{2^h | n} n^{-s} = q^h \zeta(s). \quad (9.6)$$

The finite q -binomial theorem gives the exact formal identity

$$\prod_{h=0}^{J-1} (1 + t \mathcal{Z}_h(s)) = \sum_{r=0}^J t^r \zeta(s)^r q^{\binom{r}{2}} \begin{bmatrix} J \\ r \end{bmatrix}_q. \quad (9.7)$$

This is a clean spectral explanation for Gaussian coefficients: they enumerate independent selections of dyadic zero layers, with weight q^h for choosing layer h . The identity is deliberately stated as a layer-selection polynomial; it should not be confused with a Dirichlet series for intersections, because multiplication of Dirichlet series introduces multiplicative convolution.

9.3 The common-zero zeta tower

Intersections are instead encoded by binomial moments of the multiplicity.

Theorem 9.3 (Common-zero hierarchy NEW-TO-SNAPSHOT). *For every integer $r \geq 1$ and $\Re s > 1$,*

$$Z_r(s) := \sum_{n \geq 1} \binom{a(n)}{r} n^{-s} = \zeta(s) \frac{q^{r-1}}{(1-q)^r}, \quad q = 2^{-s}. \quad (9.8)$$

For the J -layer product,

$$Z_{J,r}(s) = \sum_{n \geq 1} \binom{a_J(n)}{r} n^{-s} = \zeta(s) \sum_{h=r-1}^{J-1} \binom{h}{r-1} q^h. \quad (9.9)$$

Consequently the limiting spectrum has an order- r pole at every nonzero complex dimension χ_k unless canceled by $\zeta(\chi_k)$.

Proof. The number $\binom{a(n)}{r}$ counts r -subsets of the layers $\{0, 1, \dots, \nu_2(n)\}$. Group such subsets by their largest element h . There are $\binom{h}{r-1}$ choices and their common zeros are precisely the multiples of 2^h . Hence

$$Z_r(s) = \zeta(s) \sum_{h \geq r-1} \binom{h}{r-1} q^h = \zeta(s) \frac{q^{r-1}}{(1-q)^r}.$$

Truncating h gives the finite formula. $\square$

This tower is one of the sharpest bridges among the four themes in the user's request: the sinc layers define the zeros, the 2-adic intersection counts produce the rational powers of $(1 - q)^{-1}$, and finite layer selections produce Gaussian q -binomials.

10 The zero heat trace

Define the exponentially smoothed zero count

$$\Theta(t) = \sum_{n \geq 1} a(n) e^{-nt}, \quad t > 0. \quad (10.1)$$

Layer summation gives the dyadic Lambert series

$$\Theta(t) = \sum_{h \geq 0} \frac{1}{e^{2^h t} - 1}. \quad (10.2)$$

Its Mellin representation is

$$\Theta(t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \Gamma(s) \frac{\zeta(s)}{1-2^{-s}} t^{-s} ds, \quad c > 1. \quad (10.3)$$

Theorem 10.1 (Heat-trace expansion NEW-TO-SNAPSHOT). *Let $T = \log_2(1/t)$ and*

$$C_\Theta = \frac{\gamma - \log(2\pi)}{2 \log 2} - \frac{1}{4} = -1.1593749760977258236984308878 \dots \quad (10.4)$$

Then, as $t \downarrow 0$,

$$\begin{aligned} \Theta(t) \sim & \frac{2}{t} - \frac{\log(1/t)}{2 \log 2} + C_\Theta + \sum_{k \neq 0} \frac{\Gamma(\chi_k) \zeta(\chi_k)}{\log 2} e^{2\pi i k T} \\ & - \sum_{m=0}^{\infty} \frac{Z_\Phi(-(2m+1))}{(2m+1)!} t^{2m+1}. \end{aligned} \quad (10.5)$$

The first algebraic terms are

$$-\frac{t}{12} + \frac{t^3}{5040} + \dots \quad (10.6)$$

The first periodic coefficient has magnitude $1.2764467470886077 \times 10^{-6}$.

Proof architecture. Shift the contour in (10.3). The pole at $s = 1$ gives $2/t$. The double pole at 0 gives the logarithm and constant. The poles χ_k give the periodic term. Poles of $\Gamma(s)$ at negative even integers are canceled by trivial zeros of ζ ; the negative odd integers give the final series. $\square$

The heat trace makes the spectral dimension picture especially transparent: linear zero density, a logarithmic digital correction, and a microscopic log-periodic ripple are successive residue layers of the same Dirichlet series.

11 A base- b theory

The dyadic case belongs to a larger exact family. Fix an integer $b \geq 2$ and set

$$\Phi_b(z) = \prod_{h=0}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{b^h}\right). \quad (11.1)$$

It is the Fourier transform of the C^∞ density obtained by convolving uniform laws on intervals of half-width $1/(2b^h)$. Its support is

$$\left[-\frac{b}{2(b-1)}, \frac{b}{2(b-1)} \right]. \quad (11.2)$$

Let $\nu_b(n) = \max\{h : b^h \mid n\}$, which is meaningful even when b is composite.

Theorem 11.1 (Base- b arithmetic spectrum NEW-TO-SNAPSHOT). *The zero multiplicity at a nonzero integer is*

$$a_b(n) = 1 + \nu_b(n), \quad (11.3)$$

and

$$\sum_{n \geq 1} \frac{a_b(n)}{n^s} = \frac{\zeta(s)}{1 - b^{-s}}, \quad (11.4)$$

$$\sum_{n \leq N} a_b(n) = \sum_{h \geq 0} \left\lfloor N/b^h \right\rfloor = \frac{bN - s_b(N)}{b - 1}. \quad (11.5)$$

$$(11.6)$$

Here $s_b(N)$ is the base- b digit sum. The entire function has exponential type $\pi b/(b - 1)$ and zero density $b/(b - 1)$, in exact agreement with the Cartwright density law.

Corollary 11.2 (Even-base automatic lobe signs NEW-TO-SNAPSHOT). *If b is even and $x \notin \mathbb{Z}$, then*

$$\operatorname{sgn} \Phi_b(x) = (-1)^{s_b(\lfloor |x| \rfloor)}. \quad (11.7)$$

For arbitrary b , the sign sequence $\sigma_b(N) = (-1)^{\sum_{n \leq N} a_b(n)}$ obeys

$$\sigma_b(bN + r) = (-1)^{bN+r} \sigma_b(N), \quad 0 \leq r < b. \quad (11.8)$$

Proof. Equation (11.5) has the same proof as the binary digit-sum identity. For even b , its parity equals $s_b(N)$. The recursion follows directly from $A_b(bN + r) = bN + r + A_b(N)$. $\square$

Theorem 11.3 (Base- b Strang–Fix order NEW-TO-SNAPSHOT). *On the lattice $N^{-1}\mathbb{Z}$, the exact Strang–Fix order of the density with Fourier transform Φ_b is*

$$1 + \nu_b(N). \quad (11.9)$$

The weighted periodization and Appell reproduction theorems hold verbatim with ν_2 replaced by ν_b and the corresponding centered moments.

The Mellin calculation also persists. Let $L_b = \log b$, $\chi_{b,k} = 2\pi i k / L_b$, and $S = 2\pi y$. Then

$$\log \Phi_b(iy) = \frac{b}{2(b-1)} S - \frac{\log^2 S}{2L_b} - \frac{1}{2} \log S + C_b + \Psi_b(\log_b S) + \mathcal{O}_A(S^{-A}), \quad (11.10)$$

$$C_b = \frac{\zeta''(0) + \frac{1}{2} \log^2(2\pi) - \frac{\pi^2}{24}}{L_b} - \frac{L_b}{12}, \quad (11.11)$$

$$\Psi_b(u) = \sum_{k \neq 0} \left(-\frac{\Gamma(-\chi_{b,k}) \zeta(1 - \chi_{b,k})}{L_b} \right) e^{2\pi i k u}. \quad (11.12)$$

Thus the Gamma–zeta law is not peculiar to base two; base two is where the lobe signs become the classical Thue–Morse sequence and the exact dyadic arithmetic becomes available.

12 Numerical and symbolic verification

All numerical values in this section were computed independently from the formulas in the report with 80-decimal working precision. They are checks, not proofs.

12.1 Imaginary-axis residual

The table compares the direct convergent sum

$$H(y) = \sum_{h \geq 0} \log \frac{\sinh(\pi y/2^h)}{\pi y/2^h} \quad (12.1)$$

with (8.8), retaining 20 Fourier modes of Ψ .

y	$H(y)$	direct minus asymptotic
1	1.8186408379587714	3.4873589×10^{-6}
2	5.5707984108206301	1.0496×10^{-11}
5	20.0107450110424130	-1.0907×10^{-12}
10	47.2862093875369543	-1.0907×10^{-12}
100	594.0427656248311891	2.6185×10^{-12}

At $y = 1$ the omitted superalgebraic remainder is still visible; by $y = 2$ the error is already near the working evaluation floor.

12.2 Exact digital checks

For $N = 0, 1, \dots, 15$, the cumulative multiplicity $A(N) = 2N - s_2(N)$ has parity

$$0, 1, 1, 0, 1, 0, 0, 1, 1, 0, 0, 1, 0, 1, 1, 0,$$

which is the zero–one Thue–Morse word. Direct evaluation of 80 sinc factors at one interior point of each lobe reproduces the corresponding signs. Exact integer arithmetic also verifies (3.3), (9.9), and the recurrences (3.6) through 10^6 without exception.

12.3 A reproducible verification recipe

The main identities can be checked without evaluating up itself:

1. enumerate $a(n) = 1 + \nu_2(n)$ and compare partial sums with $2N - s_2(N)$;
2. compare $\sum_{n \leq M} a(n)n^{-s}$ with $\zeta(s)/(1 - 2^{-s})$ for $\Re s > 1$;
3. evaluate $\Phi(x)$ by the sinc product and compare its sign with $(-1)^{s_2(\lfloor x \rfloor)}$;
4. compare the finite spectral series with $\zeta(s)[J]_{2^{-s}}$;
5. evaluate $H(y)$ both by (12.1) and by the residue expansion.

These checks are sufficiently orthogonal that a normalization error in 2π , a missing initial sinc factor, or an off-by-one valuation is immediately exposed.

13 What is genuinely added to the pinned corpus

The following table separates established inputs, prior observations, and the new theorem package.

Statement	Status	Comment
Canonical product and zero order $1 + \nu_2(n)$	CORPUS	Already in the main exposition and Lean canonical-product module.
All-order endpoint asymptotics and Gamma–zeta coefficients	CORPUS	Already in the corrected small-argument study.

Statement	Status	Comment
Thue–Morse lobe sign pattern	PRIOR OBSERVATION	Recorded in the 2018 MSE question; this report gives the exact zero-count proof and lobe inversion.
One critical point per real lobe	PRIOR OBSERVATION	Essentially proved in the MSE answer; recast here through Laguerre–Polya theory and the resolvent trace.
Arithmetic spectrum package, exact counting law as zero data, spectral traces	NEW-TO-SNAPSHOT	Not found assembled in the pinned TeX corpus.
Weighted partition hierarchy and sharp order $1 + \nu_2(N)$	NEW-TO-SNAPSHOT	Degree zero is classical; the exact higher stopping law was not located in the corpus.
Appell-corrected exact polynomial reproduction	NEW-TO-SNAPSHOT	Uses existing moment-deconvolution ideas in a different, sharp lattice theorem.
Mellin transform of $\log \Phi(iy)$ and complex-dimension interpretation	NEW-TO-SNAPSHOT	Explains the already-known endpoint Fourier coefficients by zero-spectrum residues.
Finite q -integer regularization and pole-condensation profile	NEW-TO-SNAPSHOT	Connects finite sinc products to the limiting imaginary pole lattice.
Gaussian q -binomial layer-selection polynomial	NEW-TO-SNAPSHOT	A spectral-layer interpretation, distinct from existing dyadic-value q -binomials.
Common-zero zeta tower Z_r	NEW-TO-SNAPSHOT	Encodes binomial multiplicity moments and higher-order complex-dimension poles.
Heat-trace expansion	NEW-TO-SNAPSHOT	A residue expansion for exponentially smoothed zero counts.
Base- b extension of the full package	NEW-TO-SNAPSHOT	Includes digit-sum signs for even bases and exact lattice order $1 + \nu_b(N)$.

Warning 13.1 (Historical priority remains narrower than proof). A targeted search found V. A. Rvachev’s two-page 1977 paper *On approximation by means of the function* $\text{up}(x)$, followed by a substantial atomic-function approximation literature. The exact formulas of [theorems 6.1](#) and [6.4](#) were not located in the pinned corpus or in the limited searchable metadata, but a Russian-language priority audit is still needed before asserting that those facts have never appeared elsewhere.

14 Lean formalization roadmap

The results separate naturally into four dependency layers.

14.1 Layer I: elementary arithmetic

Recommended first targets are

$$\begin{aligned}
 a(n) &= \sum_h \mathbf{1}_{2^h | n}, \\
 \sum_{n \leq N} a(n) &= 2N - s_2(N), \\
 (-1)^{a(n)} &= \varepsilon_{n-1} \varepsilon_n,
 \end{aligned}$$

$$(-1)^{\sum_{n \leq N} a(n)} = \varepsilon_N.$$

The existing Thue–Morse and trailing-zero modules should make these short. They would yield the lobe-sign theorem once the canonical product’s sign changes are connected to exact orders.

14.2 Layer II: finite algebra and Dirichlet series

Finite versions should precede infinite analytic statements:

- $a_J(n) = \min(J, 1 + \nu_2(n))$;
- finite counting formula (9.3);
- finite q -integer and Gaussian-binomial polynomial identities;
- finite common-layer count (9.9).

For the infinite Dirichlet identities, prove summability on $\Re s > 1$ and use nonnegative layer rearrangement before introducing meromorphic continuation.

14.3 Layer III: exact lattice reproduction

The key reusable analytic lemma is a Poisson formula for a compactly supported C^∞ function and its polynomial multiples. Then the proof of [theorem 6.1](#) reduces to:

1. identify the Fourier coefficients of the periodization;
2. rewrite them as derivatives of Φ ;
3. annihilate nonzero coefficients with the existing exact-zero-order theorem;
4. prove failure at the first unannihilated coefficient.

The Appell theorem can be formalized coefficientwise, avoiding formal power-series asymptotics.

14.4 Layer IV: complex analysis

The Mellin and heat-trace theorems need a contour-shift framework with explicit growth estimates. A pragmatic sequence is:

1. formalize the real integral (8.4);
2. prove (8.2) only in the fundamental strip;
3. record the residue computations algebraically;
4. defer the all- A remainder until a suitable vertical-contour library is available.

Even the strip identity alone gives a valuable formally checked bridge from the zero spectrum to the endpoint Laplace transform.

15 Open frontier

Open problem 15.1 (Critical-point arithmetic). Let c_n be the unique critical point in $(n, n + 1)$. Determine asymptotic laws for $c_n - n$ and $M_n = |\Phi(c_n)|$ conditioned on $\nu_2(n), \nu_2(n + 1)$, or finite binary words around n . The logarithmic derivative (3.12) converts this into a weighted electrostatic equilibrium problem with ruler-function charges.

Open problem 15.2 (From complex dimensions to real-axis decay). The imaginary-axis periodic fluctuation is completely controlled by the poles χ_k , whereas real-axis decay splits into geometric, arithmetic, RMS, and envelope exponents. Find an operator or transfer formalism in which both sets of exponents arise from one spectrum. A direct residue argument cannot work on the real axis because zeros and near-zeros dominate.

Open problem 15.3 (Sharp approximation constants). The exact Strang–Fix order gives the first possible nonzero term in a lattice quasi-interpolant error. Compute the sharp L^∞ , L^2 , and Sobolev constants after Appell correction, and determine whether powers-of-two lattices admit an all-order Richardson scheme whose coefficients have a closed q -binomial form at base $1/4$.

Open problem 15.4 (Common-zero zeta residues). Develop explicit Laurent coefficients of $Z_r(s)$ at $s = \chi_k$. Their leading term is elementary, but lower coefficients mix derivatives of zeta with powers of $\log 2$. Determine whether these coefficients control higher-order log-periodic corrections of naturally associated spectral statistics.

Open problem 15.5 (Base- b exact arithmetic). For even b , the Fourier lobe signs are the generalized digit-sum character $(-1)^{s_b(n)}$. Develop the corresponding Fabius-type distribution function, finite box-spline formulas, exact rational values at b -adic points, and Gaussian-binomial or multinomial expressions. Composite b should display a new tension between divisibility by b^h and prime-adic arithmetic.

Open problem 15.6 (Inverse spectral characterization). Classify scale multisets $\{\lambda_h\}$ for which $\prod_h \text{sinc}(\pi z/\lambda_h)$ has an automatic or regular zero-multiplicity sequence. Conversely, characterize which nonnegative automatic sequences can occur as multiplicities of a locally uniformly convergent geometric sinc product.

16 Conclusion

The Fabius–Rvachev theory is often presented as a collection of surprising bridges: a smooth bump from infinite convolutions, rational dyadic values from q -binomials, Thue–Morse signs from finite differences, and log-periodic asymptotics from a saddle point. The arithmetic zero spectrum supplies a common organizing object.

Its summatory law integrates the ruler function into the Thue–Morse lobe signs. Its exact multiplicities turn Poisson summation into a sharp 2-adic hierarchy of polynomial reproduction. Its Dirichlet series produces the complex dimensions whose residues are the Gamma–zeta endpoint coefficients. Its finite truncations are q -integers, its layer selections are Gaussian q -binomials, and its higher intersection statistics are powers of the same geometric resolvent. None of these bridges changes the established theory; they explain why its apparently separate parts keep rediscovering the same dyadic algebra.

The most immediate formal target is [theorem 6.1](#): it is exact, strong, short once the current canonical-product theorem is available, and exposes a clean new role for $\nu_2(N)$. The most conceptually unifying analytic target is [theorem 8.2](#), which identifies the endpoint periodic fluctuation as a residue trace of the Fourier zero spectrum. Together they turn an infinite sinc product from a transform formula into an arithmetic spectral object.

A Residue calculation at the origin

For completeness, write $L = \log 2$ and expand

$$\begin{aligned}\zeta(s) &= -\frac{1}{2} - \frac{1}{2} \log(2\pi)s + \frac{1}{2} \zeta''(0)s^2 + \mathcal{O}(s^3), \\ \frac{1}{1 - e^{-Ls}} &= \frac{1}{Ls} + \frac{1}{2} + \frac{Ls}{12} + \mathcal{O}(s^3), \\ \frac{\pi}{s \sin(\pi s/2)} &= \frac{2}{s^2} + \frac{\pi^2}{12} + \mathcal{O}(s^2), \\ y^s &= 1 + s \log y + \frac{s^2 \log^2 y}{2} + \mathcal{O}(s^3).\end{aligned}$$

Multiplying and rewriting $\log y = \log S - \log(2\pi)$ gives the residue

$$-\frac{\log^2 S}{2L} - \frac{1}{2} \log S + C_2,$$

with C_2 as in (8.5). The same calculation for $\Gamma(s)Z_\Phi(s)t^{-s}$ gives the double-pole contribution

$$-\frac{\log(1/t)}{2L} + \frac{\gamma - \log(2\pi)}{2L} - \frac{1}{4}.$$

B First coefficients and moments

The numerical constants used in the report are

$$\begin{aligned}\zeta''(0) &= -2.0063564559085848512101000267299604382\dots, \\ C_2 &= -1.1090453654341866821907455547094644509\dots, \\ C_\Theta &= -1.1593749760977258236984308878499458999\dots\end{aligned}$$

For the reciprocal-moment Appell family,

$$\frac{1}{M(z)} = 1 - \frac{\mu_2}{2!}z^2 + \frac{6\mu_2^2 - \mu_4}{4!}z^4 + \dots,$$

which with $\mu_2 = 1/9$ and $\mu_4 = 19/675$ yields $6\mu_2^2 - \mu_4 = 31/675$.

C Corpus concordance

The frozen tree was read recursively. The most heavily used source families were the following paths (revision siblings are omitted from this compact list):

- docs/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex;
- docs/non-formalized-research-frontiers/non-formalized-research-frontiers.tex;
- docs/archive/standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex;
- docs/archive/research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex;
- docs/archive/research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex;
- docs/archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_Thue_Morse_Convergence_Rates.tex;
- docs/archive/research-frontiers/Repeated_Integration_and_Rvachev_Up/Repeated_Integration_and_Rvachev_Up.tex;
- docs/non-formalized-research-frontiers/drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex;
- the complete docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/ revision and audit sequence;
- primary-paper sources under docs/papers/.

The glossary, walkthrough, early notes, primary-exposition revisions, supplements, q -calculus variants, dyadic-asymptotic bridges, inverse-dyadic study, and OCR siblings were also included in the collision audit. Their principal role here was to establish chronology and prevent corrected or archived claims from being mistaken for independent corroboration.

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